Dataset - https://www.kaggle.com/datasets/utkarsharya/ecommerce-purchases

```
import pandas as pd
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
```

```
data = pd.read_csv("/content/drive/MyDrive/Datasets/Ecommerce Purchases")
```

Display Top 10 rows of the dataset

```
data.head(5)
```

	Address	Lot	AM or PM	Browser Info	Company	Credit Card	CC Exp Date	CC Security Code	CC Provider	Email	
0	16629 Pace Camp Apt. 448\nAlexisborough, NE 77...	46 in	PM	Opera/9.56.(X11; Linux x86_64; sl-SI) Presto/2...	Martinez-Herman	6011929061123406	02/20	900	JCB 16 digit	pdunlap@yahoo.com	prod d
1	9374 Jasmine Spurs Suite 508\nSouth John, TN 8...	28 rn	PM	Opera/8.93.(Windows 98; Win 9x 4.90; en-US) Pr...	Fletcher, Richards and Whitaker	3337758169645356	11/18	561	Mastercard	anthony41@reed.com	
2	Unit 0065 Box 5052\nDPO AP 27450	94 vE	PM	Mozilla/5.0 (compatible; MSIE 9.0; Windows NT ...	Simpson, Williams and Pham	675957666125	08/19	699	JCB 16 digit	amymiller@morales-harrison.com	
3	7780 Julia Fords\nNew Stacy, WA 45798	36 vm	PM	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_0 ...	Williams, Marshall and Buchanan	6011578504430710	02/24	384	Discover	brent16@olson-robinson.info	
4	23012 Munoz Drive Suite 337\nNew Cynthia, TX 5...	20 IE	AM	Opera/9.58.(X11; Linux x86_64; it-IT) Presto/2...	Brown, Watson and Andrews	6011456623207998	10/25	678	Diners Club / Carte Blanche	christopherwright@gmail.com	

Next steps: Generate code with data New interactive sheet

Display Last 10 rows of the dataset

```
data.tail(5)
```

	Address	Lot	AM or PM	Browser Info	Company	Credit Card	CC Exp Date	CC Security Code	CC Provider	Email
9995	966 Castaneda Locks\nWest Juliafurt, CO 96415	92 XI	PM	Mozilla/5.0 (Windows NT 5.1) AppleWebKit/5352 ...	Randall-Sloan	342945015358701	03/22	838	JCB 15 digit	iscott@wade-garner.com
9996	832 Curtis Dam Suite 785\nNorth Edwardburgh, T...	41 JY	AM	Mozilla/5.0 (compatible; MSIE 9.0; Windows NT ...	Hale, Collins and Wilson	210033169205009	07/25	207	JCB 16 digit	mary85@hotmail.com
9997	Unit 4434 Box 6343\nDPO AE 28026-0283	74 Zh	AM	Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10_7...	Anderson Ltd	6011539787356311	05/21	1	VISA 16 digit	tyler16@gmail.com
9998	0096 English Rest\nRoystad, IA 12457	74 cL	PM	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_8;...	Cook Inc	180003348082930	11/17	987	American Express	elizabethmoore@reid.net
9999	40674 Barrett Stravenue\nGrimesville, WI 79682	64 Hr	AM	Mozilla/5.0 (X11; Linux i686; rv:1.9.5.20) Gec...	Greene Inc	4139972901927273	02/19	302	JCB 15 digit	rachelford@vaughn.com

Check datatype of each column

```
data.dtypes
```

	0
Address	object
Lot	object
AM or PM	object
Browser Info	object
Company	object
Credit Card	int64
CC Exp Date	object
CC Security Code	int64
CC Provider	object
Email	object
Job	object
IP Address	object
Language	object
Purchase Price	float64

dtype: object

Check null values in the dataset

```
data.isnull().sum()
```

	0
Address	0
Lot	0
AM or PM	0
Browser Info	0
Company	0
Credit Card	0
CC Exp Date	0
CC Security Code	0
CC Provider	0
Email	0
Job	0
IP Address	0
Language	0
Purchase Price	0

dtype: int64

How many rows and columns are there in our dataset

```
data.columns
```

```
Index(['Address', 'Lot', 'AM or PM', 'Browser Info', 'Company', 'Credit Card',
       'CC Exp Date', 'CC Security Code', 'CC Provider', 'Email', 'Job',
       'IP Address', 'Language', 'Purchase Price'],
      dtype='object')
```

```
len(data.columns)
```

```
14
```

```
len(data)
```

```
10000
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Address           10000 non-null  object
 1   Lot               10000 non-null  object
 2   AM or PM          10000 non-null  object
 3   Browser Info      10000 non-null  object
 4   Company           10000 non-null  object
 5   Credit Card       10000 non-null  int64
 6   CC Exp Date       10000 non-null  object
 7   CC Security Code  10000 non-null  int64
 8   CC Provider       10000 non-null  object
 9   Email             10000 non-null  object
 10  Job               10000 non-null  object
 11  IP Address        10000 non-null  object
 12  Language          10000 non-null  object
 13  Purchase Price    10000 non-null  float64
dtypes: float64(1), int64(2), object(11)
memory usage: 1.1+ MB
```

Highest and Lowest Purchases

```
highest = data['Purchase Price'].max()
print(highest)
```

```
99.99
```

```
lowest = data['Purchase Price'].min()
print(lowest)
```

```
0.0
```

Average Purchase Prize

```
avg_purchase_prize =data['Purchase Price'].mean()
print(avg_purchase_prize)
```

```
50.347302
```

How many people have french 'fr' as their language?

```
len( data[data['Language']== "fr"])
```

```
1097
```

```
data[data['Language']== "fr"].count()
```

	0
Address	1097
Lot	1097
AM or PM	1097
Browser Info	1097
Company	1097
Credit Card	1097
CC Exp Date	1097
CC Security Code	1097
CC Provider	1097
Email	1097
Job	1097
IP Address	1097
Language	1097
Purchase Price	1097

dtype: int64

Job Title Contains Engineer

```
data[data['Job'].str.contains("engineer" , case= False)]
```

	Address	Lot	AM or PM	Browser Info	Company	Credit Card	CC Exp Date	CC Security Code	CC Provider	Email
1	9374 Jasmine Spurs Suite 508\nSouth John, TN 8...	28 rn	PM	Opera/8.93. (Windows 98; Win 9x 4.90; en-US) Pr...	Fletcher, Richards and Whitaker	3337758169645356	11/18	561	Mastercard	anthony41@reed.com
3	7780 Julia Fords\nNew Stacy, WA 45798	36 vm	PM	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_0 ...	Williams, Marshall and Buchanan	6011578504430710	02/24	384	Discover	brent16@olson-robinson.info
50	41159 Michael Centers\nAdamsfort, RI 37108-6674	46 Ce	PM	Mozilla/5.0 (Windows 98; Win 9x 4.90; sl-SI; r...	Wright, Williams and Mendez	4008586485908075	05/19	945	JCB 16 digit	susanvalentine@obrien.org
55	27635 Maureen Bypass Apt. 883\nSandraview, SD ...	59 LJ	AM	Mozilla/5.0 (iPod; U; CPU iPhone OS 3_3 like M...	Sims-Lyons	3158113629128344	09/19	857	VISA 16 digit	adkinsarthur@yahoo.com
60	7126 Katherine Squares\nPerkinsview, CO 97299-...	63 qu	AM	Opera/8.68. (X11; Linux x86_64; en-US) Presto/2...	Marshall-Fernandez	349767747049645	05/20	672	JCB 15 digit	sweeneyhannah@jones.biz
...	...	...	...	...	...	...	...	...	...	...
9948	95544 Johnson Isle Suite 939\nMichaelberg, RI ...	91 bW	AM	Opera/8.36. (X11; Linux x86_64; sl-SI) Presto/2...	Fox-Peterson	4762924304307	03/17	567	Mastercard	haleybenjamin@gmail.com
9952	9991 Vaughn Hills\nRacheltown, PA 55409	36 KC	PM	Mozilla/5.0 (X11; Linux i686; rv:1.9.5.20) Gec...	Ward, Smith and Castillo	6011679271321726	09/19	964	Voyager	jonesjennifer@olson.com
9970	0060 Keith Stream\nWestport, CO 47097	11 nt	PM	Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10_8...	Carpenter, Good and Hart	6011485664704662	07/19	543	Discover	rangelbrian@hotmail.com
9977	02182 Keith Expressway\nEast Shannon, CT 20578...	34 RL	AM	Mozilla/5.0 (compatible; MSIE 9.0; Windows NT ...	Deleon, Jacobson and Benton	4186094003664688	06/21	397	JCB 16 digit	daltoncarter@yahoo.com
9996	832 Curtis Dam Suite 785\nNorth Edwardburgh, T...	41 JY	AM	Mozilla/5.0 (compatible; MSIE 9.0; Windows NT ...	Hale, Collins and Wilson	210033169205009	07/25	207	JCB 16 digit	mary85@hotmail.com

984 rows × 14 columns

Find the email of the person with the following ip address: 132.207.106.22

```
data[data['IP Address']=="132.207.160.22"]
```

	Address	Lot	AM or PM	Browser Info	Company	Credit Card	CC Exp Date	CC Security Code	CC Provider	Email	Job	IP Address
2	Unit 0065 Box 5052\nDPO AP 27450	94 vE	PM	Mozilla/5.0 (compatible; MSIE 9.0; Windows	Simpson, Williams and Pham	675957666125	08/19	699	JCB 16 digit	amymiller@morales-harrison.com	Customer service manager	132.207.160.22

```
data[data['IP Address']=="132.207.160.22"]['Email']
```

	Email
2	amymiller@morales-harrison.com

dtype: object

How many people have Mastercard as their credit card provider and made purchase above 50 ?

```
data[(data['CC Provider']=="Mastercard" ) & (data['Purchase Price']>50 ) ]
```

```
len(data[(data['CC Provider']=="Mastercard" ) & (data['Purchase Price']>50 ) ])
```

```
405
```

```
data[(data['CC Provider']=="Mastercard" ) & (data['Purchase Price']>50 ) ].count()
```

	0
Address	405
Lot	405
AM or PM	405
Browser Info	405
Company	405
Credit Card	405
CC Exp Date	405
CC Security Code	405
CC Provider	405
Email	405
Job	405
IP Address	405
Language	405
Purchase Price	405

dtype: int64

Find Email of the person with the following Credit Card Number: 4664825258997302

```
data[data['Credit Card']==4664825258997302]['Email']
```

	Email
9992	bberry@wright.net

dtype: object

How many people purchase during the AM and how many people purchase during the PM

```
len(data[data['AM or PM']=="AM"])
```

```
4932
```

```
len(data[data['AM or PM']=="PM"])
```

```
5068
```

```
data['AM or PM'].value_counts()
```

AM or PM	count
PM	5068
AM	4932

dtype: int64

How many people have a credit that expires in 2020

```
exp_2020 = data[data['CC Exp Date'].str[-2:] == '20']
exp_2020
```

```
count_2020 = exp_2020.shape[0]
count_2020
```

```
988
```

```
def func():
  c=0
  for date in data['CC Exp Date']:
    if date.split('/')[1]=='20':
      c+=1
  return c
func()
```

```
988
```

```
len(data[data['CC Exp Date'].apply(lambda x:x[3:]=="20")])
```

```
988
```

Top 5 most popular email providers (eg. gmail.com, yahoo.com, etc...)

```
data['Email'].str.split('@')
```

	Email
0	[pdunlap, yahoo.com]
1	[anthony41, reed.com]
2	[amymiller, morales-harrison.com]
3	[brent16, olson-robinson.info]
4	[christopherwright, gmail.com]
...	...
9995	[iscott, wade-garner.com]
9996	[mary85, hotmail.com]
9997	[tyler16, gmail.com]
9998	[elizabethmoore, reid.net]
9999	[rachelford, vaughn.com]

10000 rows × 1 columns

dtype: object

```
l=data['Email'].apply(lambda x:x.split('@')[1]).value_counts().head()
l
```

Email	count
hotmail.com	1638
yahoo.com	1616
gmail.com	1605
smith.com	42
williams.com	37

dtype: int64

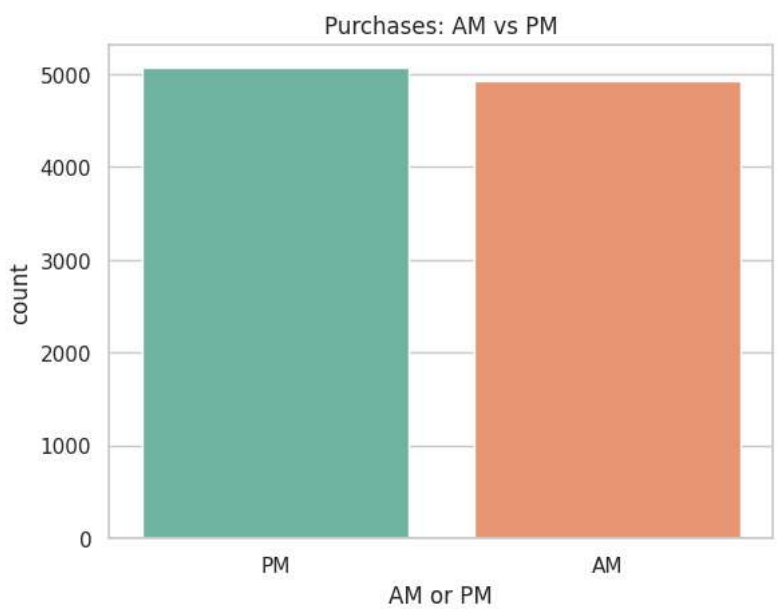
E-Commerce Purchases by Time of Day (AM vs PM)

```
ampm = data['AM or PM'].value_counts()
ampm
```

AM or PM	count
PM	5068
AM	4932

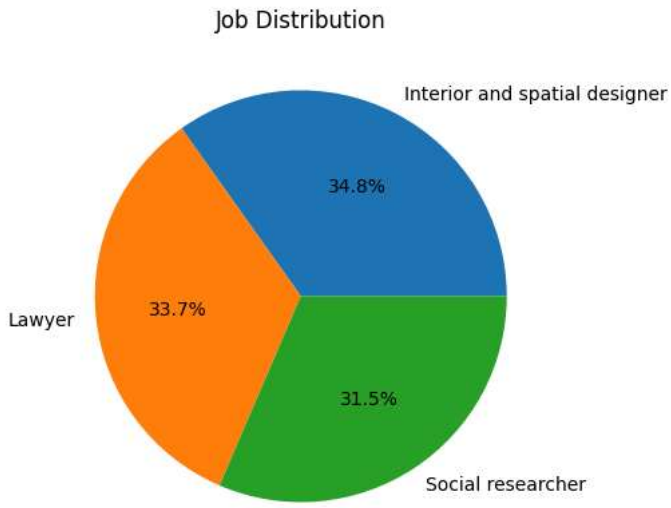
dtype: int64

```
sns.countplot(x='AM or PM', data=data, palette='Set2' ,   hue='AM or PM')
plt.title("Purchases: AM vs PM")
plt.show()
```



Top Professions Among E-Commerce Customers

```
lang = data['Job'].value_counts().head(3)

plt.figure(figsize=(6,5))
plt.pie(lang, labels=lang.index, autopct='%1.1f%%')
plt.title('Job Distribution')
plt.show()
```



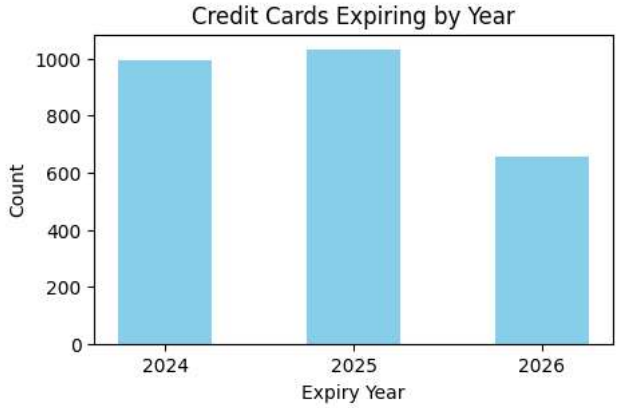
Total Credit card expiry in next 3 years

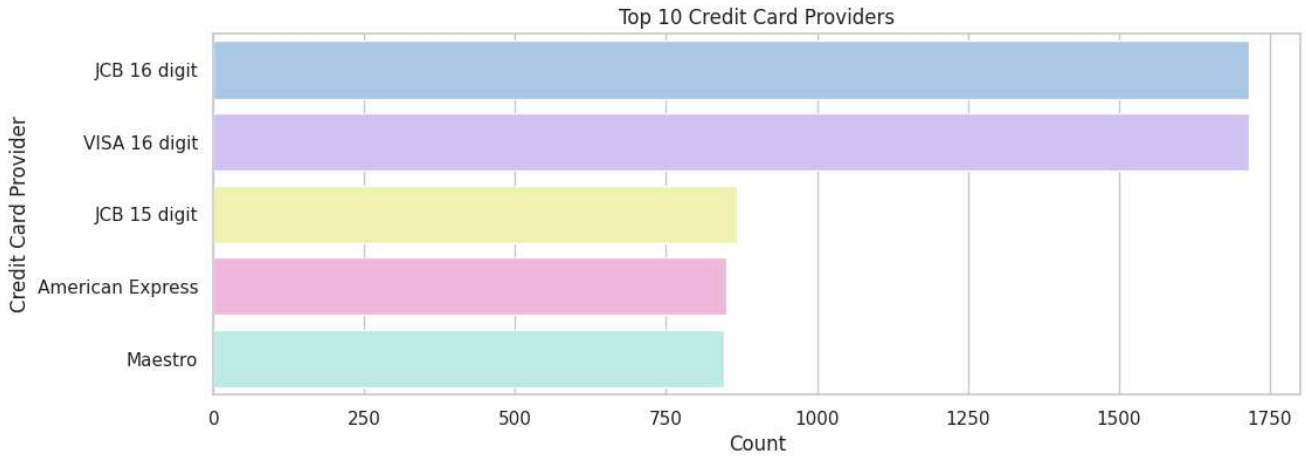
```
exp_24 =len(data[data['CC Exp Date'].apply(lambda x:x[3:]=="24")])
exp_25 =len(data[data['CC Exp Date'].apply(lambda x:x[3:]=="25")])
exp_26 =len(data[data['CC Exp Date'].apply(lambda x:x[3:]=="26")])
print(exp_24,exp_25,exp_26)
```

```
992 1033 654
```

```
years = ['2024', '2025', '2026']
values = [exp_24, exp_25, exp_26]

plt.figure(figsize=(5,3))
plt.bar(years, values, color='skyblue', width=0.5)

plt.title('C', fontsize=12)
plt.xlabel('Expiry Year', fontsize=10)
plt.ylabel('Count', fontsize=10)
```

```
Text(0, 0.5, 'Count')
```




```
import seaborn as sns
import matplotlib.pyplot as plt
plt.figure(figsize=(12,4))

ax = sns.countplot(
    y='CC Provider',
    data=data,
    order=data['CC Provider'].value_counts().index[:5],
    hue='CC Provider',        # required when using palette
    palette='pastel',
    legend=False              # hide legend
)
```

```
)

plt.title("Top 10 Credit Card Providers")
plt.xlabel("Count")
plt.ylabel("Credit Card Provider")
plt.show()
```